

Ejercicio 2

Trace Pc1 a Pc4:

```
PC1> trace 192.24.0.130
trace to 192.24.0.130, 8 hops max, press Ctrl+C to stop
 1  192.24.0.1   15.677 ms  16.416 ms  16.030 ms
 2  192.24.0.157 48.115 ms  48.135 ms  48.141 ms
 3  192.24.0.153 79.411 ms  80.126 ms  80.239 ms
 4  *192.24.0.130 96.000 ms (ICMP type:3, code:3, Destination port unreachable)
```

R1-00401624:

```
R1-00401624#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          192.24.0.129    YES manual up          up
FastEthernet1/0          192.24.0.153    YES manual up          up
Serial2/0                 192.24.0.162    YES manual up          up
Serial2/1                 unassigned      YES unset  administratively down down
Serial2/2                 unassigned      YES unset  administratively down down
Serial2/3                 unassigned      YES unset  administratively down down

R1-00401624#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    192.24.0.0/24 is variably subnetted, 10 subnets, 4 masks
C       192.24.0.152/30 is directly connected, FastEthernet1/0
O       192.24.0.156/30 [110/65] via 192.24.0.154, 00:11:39, FastEthernet1/0
O       192.24.0.144/29 [110/129] via 192.24.0.161, 00:09:19, Serial2/0
C       192.24.0.128/28 is directly connected, FastEthernet0/0
O       192.24.0.168/30 [110/129] via 192.24.0.154, 00:11:39, FastEthernet1/0
C       192.24.0.160/30 is directly connected, Serial2/0
O       192.24.0.164/30 [110/128] via 192.24.0.161, 00:10:59, Serial2/0
O       192.24.0.64/27 [110/2] via 192.24.0.154, 00:11:39, FastEthernet1/0
O       192.24.0.96/27 [110/65] via 192.24.0.161, 00:10:59, Serial2/0
O       192.24.0.0/27 [110/66] via 192.24.0.154, 00:11:39, FastEthernet1/0
```

R2-00401624:

```

R2-00401624#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          192.24.0.65     YES manual up           up
FastEthernet1/0          192.24.0.154    YES manual up           up
Serial2/0                 192.24.0.157    YES manual up           up
Serial2/1                 unassigned      YES unset  administratively down down
Serial2/2                 unassigned      YES unset  administratively down down
Serial2/3                 unassigned      YES unset  administratively down down
R2-00401624#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    192.24.0.0/24 is variably subnetted, 10 subnets, 4 masks
C       192.24.0.152/30 is directly connected, FastEthernet1/0
C       192.24.0.156/30 is directly connected, Serial2/0
O       192.24.0.144/29 [110/129] via 192.24.0.158, 00:09:55, Serial2/0
O       192.24.0.128/28 [110/2] via 192.24.0.153, 00:12:31, FastEthernet1/0
O       192.24.0.168/30 [110/128] via 192.24.0.158, 00:16:39, Serial2/0
O       192.24.0.160/30 [110/65] via 192.24.0.153, 00:11:45, FastEthernet1/0
O       192.24.0.164/30 [110/129] via 192.24.0.153, 00:11:35, FastEthernet1/0
C       192.24.0.64/27 is directly connected, FastEthernet0/0
O       192.24.0.96/27 [110/66] via 192.24.0.153, 00:11:35, FastEthernet1/0
O       192.24.0.0/27 [110/65] via 192.24.0.158, 00:16:39, Serial2/0

```

R3-00401624:

```

R3-00401624#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          192.24.0.97     YES manual up          up
FastEthernet1/0          unassigned      YES unset  administratively down down
Serial2/0                 192.24.0.161    YES manual up          up
Serial2/1                 192.24.0.166    YES manual up          up
Serial2/2                 unassigned      YES unset  administratively down down
Serial2/3                 unassigned      YES unset  administratively down down
R3-00401624#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    192.24.0.0/24 is variably subnetted, 10 subnets, 4 masks
O       192.24.0.152/30 [110/65] via 192.24.0.162, 00:12:33, Serial2/0
O       192.24.0.156/30 [110/129] via 192.24.0.162, 00:12:33, Serial2/0
O       192.24.0.144/29 [110/65] via 192.24.0.165, 00:11:03, Serial2/1
O       192.24.0.128/28 [110/65] via 192.24.0.162, 00:12:33, Serial2/0
O       192.24.0.168/30 [110/128] via 192.24.0.165, 00:10:39, Serial2/1
C       192.24.0.160/30 is directly connected, Serial2/0
C       192.24.0.164/30 is directly connected, Serial2/1
O       192.24.0.64/27 [110/66] via 192.24.0.162, 00:12:33, Serial2/0
C       192.24.0.96/27 is directly connected, FastEthernet0/0
O       192.24.0.0/27 [110/129] via 192.24.0.165, 00:10:39, Serial2/1

```

R4-00401624:

```

R4-00401624#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          192.24.0.145    YES manual up          up
FastEthernet1/0          unassigned      YES unset   administratively down down
Serial2/0                 192.24.0.165    YES manual up          up
Serial2/1                 192.24.0.170    YES manual up          up
Serial2/2                 unassigned      YES unset   administratively down down
Serial2/3                 unassigned      YES unset   administratively down down
R4-00401624#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    192.24.0.0/24 is variably subnetted, 10 subnets, 4 masks
O       192.24.0.152/30 [110/129] via 192.24.0.169, 00:11:28, Serial2/1
        [110/129] via 192.24.0.166, 00:11:52, Serial2/0
O       192.24.0.156/30 [110/128] via 192.24.0.169, 00:11:28, Serial2/1
C       192.24.0.144/29 is directly connected, FastEthernet0/0
O       192.24.0.128/28 [110/129] via 192.24.0.166, 00:11:52, Serial2/0
C       192.24.0.168/30 is directly connected, Serial2/1
O       192.24.0.160/30 [110/128] via 192.24.0.166, 00:11:52, Serial2/0
C       192.24.0.164/30 is directly connected, Serial2/0
O       192.24.0.64/27 [110/129] via 192.24.0.169, 00:11:28, Serial2/1
O       192.24.0.96/27 [110/65] via 192.24.0.166, 00:11:52, Serial2/0
O       192.24.0.0/27 [110/65] via 192.24.0.169, 00:11:28, Serial2/1

```

R5-00401624:

```

R5-00401624#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    192.24.0.0/24 is variably subnetted, 10 subnets, 4 masks
O       192.24.0.152/30 [110/65] via 192.24.0.157, 00:19:07, Serial2/0
C       192.24.0.156/30 is directly connected, Serial2/0
O       192.24.0.144/29 [110/65] via 192.24.0.170, 00:12:18, Serial2/1
O       192.24.0.128/28 [110/66] via 192.24.0.157, 00:14:59, Serial2/0
C       192.24.0.168/30 is directly connected, Serial2/1
O       192.24.0.160/30 [110/129] via 192.24.0.157, 00:14:13, Serial2/0
O       192.24.0.164/30 [110/128] via 192.24.0.170, 00:12:18, Serial2/1
O       192.24.0.64/27 [110/65] via 192.24.0.157, 00:19:07, Serial2/0
O       192.24.0.96/27 [110/129] via 192.24.0.170, 00:12:18, Serial2/1
C       192.24.0.0/27 is directly connected, FastEthernet0/0
R5-00401624#

```

PC1:

```
PC1> show ip

NAME       : PC1[1]
IP/MASK    : 192.24.0.2/26
GATEWAY    : 192.24.0.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 10050
RHOST:PORT : 127.0.0.1:10051
MTU       : 1500
```

PC2:

```
PC2> show ip

NAME       : PC2[1]
IP/MASK    : 192.24.0.66/27
GATEWAY    : 192.24.0.65
DNS        :
MAC        : 00:50:79:66:68:01
LPORT     : 10052
RHOST:PORT : 127.0.0.1:10053
MTU       : 1500
```

PC3:

```
PC3> show ip

NAME       : PC3[1]
IP/MASK    : 192.24.0.98/27
GATEWAY    : 192.24.0.97
DNS        :
MAC        : 00:50:79:66:68:02
LPORT     : 10054
RHOST:PORT : 127.0.0.1:10055
MTU       : 1500
```

PC4:

```
PC4> show ip

NAME       : PC4[1]
IP/MASK    : 192.24.0.130/28
GATEWAY    : 192.24.0.129
DNS        :
MAC        : 00:50:79:66:68:03
LPORT     : 10056
RHOST:PORT : 127.0.0.1:10057
MTU       : 1500
```

PC5:

```
PC5> show ip

NAME       : PC5[1]
IP/MASK    : 192.24.0.146/29
GATEWAY    : 192.24.0.145
DNS        :
MAC        : 00:50:79:66:68:04
LPORT      : 10058
RHOST:PORT : 127.0.0.1:10059
MTU        : 1500
```

Ejercicio 1:

PC1:

```
PC1> show ip

NAME       : PC1[1]
IP/MASK    : 10.24.0.2/26
GATEWAY    : 10.24.0.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 10028
RHOST:PORT : 127.0.0.1:10029
MTU        : 1500
```

PC2:

```
PC2> show ip

NAME       : PC2[1]
IP/MASK    : 10.24.0.66/27
GATEWAY    : 10.24.0.65
DNS        :
MAC        : 00:50:79:66:68:01
LPORT      : 10032
RHOST:PORT : 127.0.0.1:10033
MTU        : 1500
```

PC3:

```
PC3> show ip

NAME       : PC3[1]
IP/MASK    : 10.24.0.98/28
GATEWAY    : 10.24.0.97
DNS        :
MAC        : 00:50:79:66:68:02
LPORT     : 10030
RHOST:PORT : 127.0.0.1:10031
MTU        : 1500
```

PC4:

```
PC4> show ip

NAME       : PC4[1]
IP/MASK    : 10.24.0.114/28
GATEWAY    : 10.24.0.113
DNS        :
MAC        : 00:50:79:66:68:03
LPORT     : 10034
RHOST:PORT : 127.0.0.1:10035
MTU        : 1500
```

R1-00401624:

```
R1-00401624#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 10.24.0.1       YES manual up          up
FastEthernet1/0 10.24.0.65      YES manual up          up
Serial2/0       10.24.0.129     YES manual up          up
Serial2/1       unassigned      YES unset  administratively down down
Serial2/2       unassigned      YES unset  administratively down down
Serial2/3       unassigned      YES unset  administratively down down
R1-00401624#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 3 subnets, 3 masks
C    10.24.0.0/26 is directly connected, FastEthernet0/0
C    10.24.0.64/27 is directly connected, FastEthernet1/0
C    10.24.0.128/30 is directly connected, Serial2/0
```

R2:

```
R2-00401624#show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          10.24.0.97      YES manual up              up
FastEthernet1/0          unassigned      YES unset  administratively down down
Serial2/0                 10.24.0.130     YES manual up              up
Serial2/1                 10.24.0.133     YES manual up              up
Serial2/2                 unassigned      YES unset  administratively down down
Serial2/3                 unassigned      YES unset  administratively down down
R2-00401624#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.24.0.96/28 is directly connected, FastEthernet0/0
C       10.24.0.128/30 is directly connected, Serial2/0
C       10.24.0.132/30 is directly connected, Serial2/1
R2-00401624#
```

R3:

```
R3-00401624#show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          10.24.0.113     YES manual up              up
FastEthernet1/0          unassigned      YES unset  administratively down down
Serial2/0                 10.24.0.134     YES manual up              up
Serial2/1                 unassigned      YES unset  administratively down down
Serial2/2                 unassigned      YES unset  administratively down down
Serial2/3                 unassigned      YES unset  administratively down down
R3-00401624#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.24.0.112/28 is directly connected, FastEthernet0/0
C       10.24.0.132/30 is directly connected, Serial2/0
R3-00401624#
```

Trace pc4 a pc1:

```
PC4> traceroute 10.24.0.1
Bad command: "traceroute 10.24.0.1". Use ? for help.

PC4> trace 10.24.0.1
trace to 10.24.0.1, 8 hops max, press Ctrl+C to stop
 1  10.24.0.113   16.632 ms  16.342 ms  15.899 ms
 2  *10.24.0.113  15.958 ms (ICMP type:3, code:1, Destination host unreachable)
)
```